

MUHAMMAD HAZIQ HASBI BIN OSMAN

haziqhasbi559@gmail.com | +60194157795 | Butterworth, Penang



PROFILE

Results-driven Electronic Engineering graduate with practical experience in machine maintenance, electronic component testing, and network system installation. Completed internships at Robert Bosch Semiconductor Manufacturing and SSIS E Solution. Skilled in soldering, troubleshooting, supporting spare part management, machine buy-off, and workpiece carrier repair and data tracking using Microsoft Excel. Strong understanding of industrial systems, equipment reliability, and electronic hardware functionality, with a commitment to quality and continuous improvement.

EDUCATION

Universiti Tun Hussein Onn Malaysia (UTHM)

Bachelor's Degree with Honours in Electronic Engineering

Batu Pahat, Johor

(September 2021 – July 2025)

Key Coursework: Embedded Systems, Microprocessor & Microcontroller, Analog & Digital Electronics, Computer Programming, Artificial Intelligence, Digital Signal Processing.

Politeknik Seberang Perai

Diploma in Electrical & Electronic Engineering

Permatang Pauh, Penang

(Jun 2018 – August 2021)

Key Coursework: Embedded Robotics, PLC, Semiconductor Devices, Electronic Circuits, Communication Systems, Equipment Repair.

WORK EXPERIENCED

Diploma Internship

SSIS E Solution Sdn. Bhd.

Bukit Mertajam, Penang

(March 2021 – July 2021)

- Assisted in the installation and configuration of fiber optic networks and server racks, ensuring compliance with network layout and safety standards.
- Performed Ethernet cable termination, routing, and labeling for LAN infrastructure to support high-speed data connectivity.
- Supported network equipment setup including switches, routers, and patch panels for new client installations.
- Performed signal testing, troubleshooting, and cable certification using network diagnostic tools to ensure optimal link performance.
- Maintained installation logs and updated network layout documentation for project delivery and audit compliance.

Degree Internship

Robert Bosch Semiconductor Manufacturing Penang Sdn. Bhd.

Batu Kawan, Penang

(July 2024 – September 2024)

- Developed an Excel-based spare parts tracking system using table listings to improve inventory management and tracking.
- Conducted testing and repair of workpiece carriers, identifying defective components and performing root cause analysis to restore functionality.
- Performed soldering and component replacement on PCB assemblies, ensuring compliance with ESD and IPC standards for rework quality.

IC Design Training Program: Design Verification

Penang Skill Development Center (PSDC)

Bayan Lepas, Pulau Pinang

(November 2025 – December 2025)

- Hands-on experience developing and running SystemVerilog and Verilog testbenches, including simulation execution, waveform analysis, and debugging in a Linux-based environment
- Gained fundamental understanding of CMOS technology, including basic transistor operation, logic gates, and their role in digital IC design
- Developed a solid understanding of UVM (Universal Verification Methodology) concepts and verification flow
- Practiced writing SystemVerilog Assertions (SVA) to verify functional behavior and timing conditions
- Implemented basic UVM components such as sequences, drivers, monitors, and scoreboards
- Applied assertion-based and simulation-based verification techniques in lab exercises

ACADEMIC PROJECT

Prediction For Material Health Using Tree Based Machine Learning Model

Johor

Degree's Final Year Project

(2025)

- Developed a machine learning model to predict maternal health risk levels (low, medium, high) based on clinical attributes such as blood pressure, blood sugar, and heart rate.

Energy Consumption Monitoring System with IOT

Johor

Degree's Integrated Design Project

(2024)

- Designed and developed an IoT-based system to monitor and analyze real-time energy consumption for residential and small industrial environments, enabling remote tracking to optimize energy usage and reduce costs.

Autonomous Fire Fighter Robot

Penang

Diploma's Final Year Project

(2021)

- Developed an autonomous fire-fighting robot to detect and extinguish small indoor fires without human intervention. The robot navigates autonomously, identifies fire sources using sensors, and activates a suppression mechanism for early fire control.

CERTIFICATE AND TRAINING

Integrated Design Project (IDP) Showcase - UTHM Silver Award	Johor (2024)
Training Matlab Calculus and Ordinary Differential Equation	Johor (2024)
SULAM Project - FKEE	Johor (2023)
Production Workshop Program – UTHM Multimedia Club	Johor (2021)
Dean’s List Award Sem 2 & 7	Johor (2025)
IC Design Training Program: Design Verification - PSDC	Pulau Pinang (2025)

SKILLS

Technical Skills: Microsoft Office, LINUX, Microsoft Word, Microsoft Excel, Microsoft Teams, C ++ Programming, Python, MATLAB, Software Tools (Multisim, Arduino IDE, Proteus, Vivado, Xilinx SDK).

Practical Skills: Assembly Product, Preventive Maintenance, Buy-off Machine, Soldering, Basic Troubleshooting.

Language: Malay, English

REFERENCES

DR. NUR ILYANI BINTI RAMLI Supervisor Final Year Project UTHM 013-4549494	ANBARASU BALASUBRAMANIAM Supervisor Robert Bosch Semiconductor Manufacturing 016-4081259
--	---